ACCCALC DOCUMENTATION

AccCalc v9.0.0

This handoff package documents the final `9.0.0` state of AccCalc after the academic-depth release that rebuilt Study Hub into a textbook-style lesson reader and upgraded OCR plus Smart Solver parsing for richer accounting problems.

Release Summary

- Rebuilt Study Hub around a reusable lesson reader layout.
- Expanded reviewer coverage across FAR, AFAR, audit, tax, RFBT, AIS, governance, operations, and strategic integration.
- Added a shared number-parsing utility for Smart Solver and OCR.
- Improved OCR cleanup, structured fields, and review-needed guidance.
- Fixed Study Hub track navigation so breadcrumbs return to the right shelf.

Architecture Overview

Routes

Route registration lives in `src/App.tsx` and points to lazy-loaded feature pages.

Catalog

`src/utils/appCatalog.ts` is the route metadata source of truth for search, navigation, and curriculum grouping.

Study

`studyContent.ts`, `studyExpansion450.ts`, and the

OCR & Smart Solver

`ocrParser.ts`, `ocrMathCleanup.ts`, and

new lesson layout define the learning system.

`smartSolver.engine.ts` now share safer numeric parsing.

Study Hub Lesson System

The new lesson architecture centers on `StudyLessonLayout.tsx`. It provides a consistent reader shell with breadcrumbs, outline, active section tracking, progress, resume state, and related-learning panels.

> The lesson experience now behaves more like a digital accounting textbook chapter than a stack of independent dashboard cards.

Academic Expansion Areas

- FAR refinements for financial assets, provisions, and unearned revenue.
- AFAR partnership life-cycle coverage from admission through final exit.
- Audit linkage between materiality, risk, assertions, and evidence.
- Tax compliance depth with withholding, local tax, remedies, and compliance posture.
- RFBT modules for obligations and contracts plus corporation law and liquidation.
- AIS, governance, operations, and strategic integration reviewer expansion.

OCR And Structured Review

Scan review is now more transparent. Structured fields can keep a raw value, normalized value, value kind, source line, confidence, and a review-needed marker before the user trusts autofill or route suggestions.

Cleanup

`ocrMathCleanup.ts` normalizes spacing, currencies, percentages, and line structure.

Parsing

`ocrParser.ts` classifies the text, extracts candidate values, and creates structured review fields.

Accounting Sheets

`accountingFieldExtractor.ts` handles worksheet-like process-costing pages with field-level metadata.

Review UI

`ScanStructuredFieldsEditor.tsx` surfaces confidence and source-line context instead of raw text only.

Validation

The final release state passed:

- `npm test`
- `npm run build`

Known Remaining Limits

- Large production chunks still remain in the workpaper, catalog, solve-definition, and main-shell surfaces.
- OCR remains a guided review flow rather than guaranteed-perfect extraction.
- Several new curriculum areas in this release expand through lesson modules more than through brand-new standalone calculators.